MASTERING D AND F BLOCK ELEMENTS FOR JEE

INTRODUCTION TO D AND F BLOCK ELEMENTS

1. Main Concept
• D-block: Last electron enters the d-subshell of the penultimate shell (n−1).
• F-block: Last electron enters the f-subshell of the anti-penultimate shell (n−2).
• Position: D-block (Groups 3 to 12); F-block (Separate rows at the bottom).

2. Series of Elements
• 3d Series (First Transition): Sc (Z=21) to Zn (Z=30). Most important for JEE.
Sono tum vyah karlo mujse phir koi nahi kahega zanu
Sc Ti V Cr Mn Fe Co Ni Cu Zn
• 4d Series (Second Transition): Y (Z=39) to Cd (Z=48).
Yaari zara nibha moat tak rukavat rah *m* pade aag *m* kud ja
Y Zr Nb Mo Tc Ru Rh Pd Ag Cd
• 5d Series (Third Transition): La (Z=57) to Hg (Z=80).
La haf'ta warna re odhar idhar pitai aur hogi
La Hf Ta W Re Os Ir Pt Au Hg
• 6d Series (Fourth Transition): Starts with Ac (Z=89).
• Lanthanoids (4f): Ce (Z=58) to Lu (Z=71).
• Actinoids (5f): Th (Z=90) to Lr (Z=103).

coinage metal : Cu, Ag, Au
Volatile metal : Zn, Cd, Hg

3. Key Concepts
• Inner Transition Elements: F-block elements are known as inner transition elements.
• Syllabus Focus: Electronic configurations, oxidation states, and Lanthanoid/Actinoid contractions are high-priority.

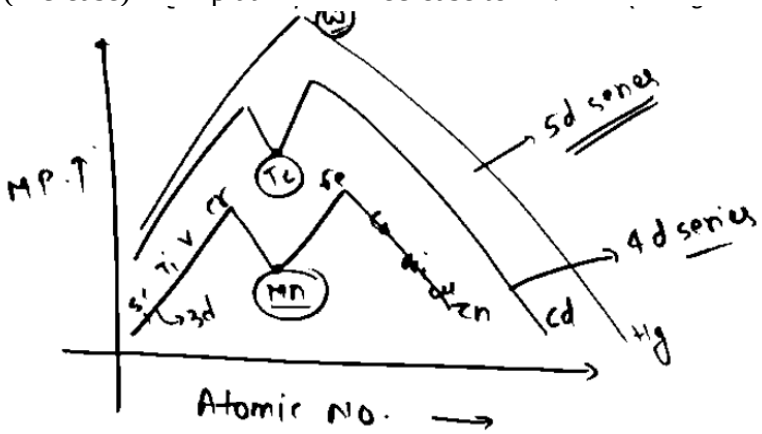
D-BLOCK PROPERTIES AND CHARACTERISTICS

1. General Electronic Configuration
• (Formula) General d-block: $(n-1)d^{1-10}\ ns^{0-2}$
• Palladium (Z=46) Exception: $4d^{10}\ 5s^{0}$ (Only d-block element with zero s-electrons).
• Stability rule: Half-filled (d^5) and full-filled (d^{10}) configurations are extra stable.
• Cr (Z=24): [Ar] $3d^5\ 4s^1$
• Cu (Z=29): [Ar] $3d^{10}\ 4s^1$
• (Mnemonic for Exceptions): "Nawabi kyun karein maut tak, rukawat raah mein padegi, aage pitayi aur hogi" (Nb, Cu, Cr, Mo, Tc?, Ru, Rh, Pd, Ag, Pt, Au). *# sirf ek electron aage peche*

2. Transition Elements Definition
• Definition: Elements with an incompletely filled d-orbital in ground state OR any stable oxidation state.
• Important Exception: Zn, Cd, and Hg are d-block elements but NOT transition elements because they have d^{10} configurations in ground and stable (+2) states.
• Sc (Z=21): Transition element ($3d^1$ ground state).
• Cu (Z=29): Transition element (Cu^{2+} is $3d^9$).

3. Physical Properties and Hardness
• Typical Metals: High tensile strength, malleability, ductility, high thermal and electrical conductivity and luster.
• Exceptions: Zn, Cd, and Hg are soft. Hg is liquid at room temperature.
• Metallic Structures: BCC, HCP, or CCP. Zn, Cd, Hg, and Mn show exceptions in typical structures.

4. Melting Point (MP) Trends
• Cause of high MP: Strong metallic bonding involving both ns and (n−1)d electrons.
• Bond Strength ∝ Number of unpaired electrons.
• Bond Strength ∝ Z_eff (Effective Nuclear Charge).
• (Note) Mn Anomaly: Mn ($3d^5$) has an exceptionally low MP because its e^- are very stable in half-filled shells and participate less in bonding.
• 3d Trend: Sc to Cr (Increase) → Dip at Mn → Decrease to Zn.



• Group Trend: 5d > 4d > 3d (Higher Z_eff in 5d means stronger metal-metal bonding).
• Highest MP: Tungsten (W).
• Lowest MP: Mercury (Hg).

5. Enthalpy of Atomization (ΔH_a)
They have high Enthalpy of Atomization.
• Definition: Energy to convert 1 mole of solid metal to gaseous atoms.
• (Formula) M(s) → M(g)
• Trend: Directly proportional to metallic bond strength/unpaired electrons.
• Max value: Middle of the series (maximum unpaired e^-).
• Order: 5d > 4d > 3d.
• Lowest ΔH_a in d-block: Zinc (Zn) due to zero unpaired electrons.

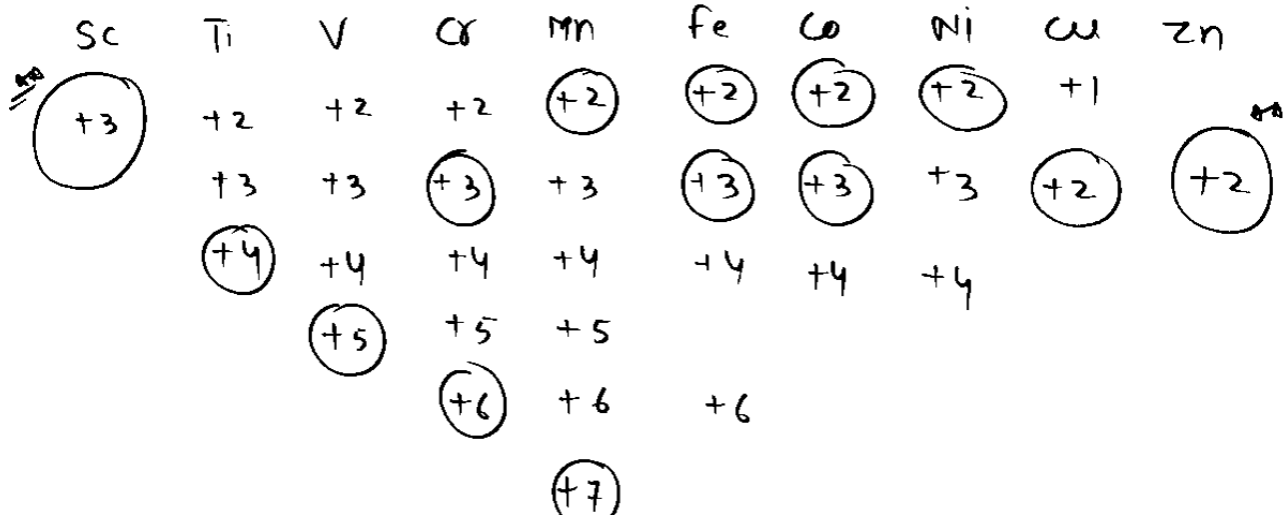
6. Atomic and Ionic Radii
• 3d Trend (Across): Sc to Mn (Decrease due to Z_eff) → Fe to Ni (Constant due to shielding) → Cu to Zn (Increase due to e^-−e^- repulsion).
• Group Trend: 3d < 4d ≈ 5d.
• **Lanthanoid Contraction: 4f electrons shield poorly, allowing the nucleus to pull outer shells strongly. This makes Zr (4d) and Hf (5d) similar in size.**
• (Exception) Group 3: Size increases normally (3d < 4d < 5d).

7. Density and Ionization Enthalpy (IE)
• Density = Mass / Volume.
• Trend: Increases across series. Max at Cu, dip at Zn (due to Zn's larger volume).
• IE ∝ 1 / Radius.
• Group 3 IE: 3d > 4d > 5d.
• **Groups 4-12 IE: 5d > 3d > 4d** (High 5d IE due to Lanthanoid contraction/Z_eff).

• Vanadium Anomaly: IE of V < Ti due to Exchange Energy released during d^3 stability.
• Successive IE: High IE_3 for Mn and Zn prevents +3 oxidation state.

8. Magnetic Properties
• (Formula) Spin-only magnetic moment: $\mu = \sqrt{[n(n+2)]}$ BM
• n = number of unpaired electrons; BM = Bohr Magnetons.
• Paramagnetic: ≥ 1 unpaired e^-.
• Diamagnetic: 0 unpaired e^-.

9. Oxidation States
• Variability: Due to small energy gap between ns and (n−1)d orbitals.
• Scandium: +3 only (most stable).
• Zinc: +2 only (most stable).
• Manganese: Max states in 3d (+2 to +7).
• Max in d-block: Ru and Os (+8).



• **D-block vs P-block: Higher oxidation states are more stable DOWN the group in d-block (e.g., Mo^{VI} and W^{VI} are more stable than Cr^{VI}).**
• Stabilization: F and O stabilize high states (*Oxygen is superior due to multiple bonding*).

D-BLOCK CHARACTERISTICS AND TRENDS

1. Alloy and Interstitial Compounds
• Alloy: Homogeneous solid solution of metals. Constituents must have radii within 15%.
• Transition metal alloys: Hard, high MP (*e.g., Brass: Cu-Zn, Bronze: Cu-Sn*).
• Interstitial Compounds: Small atoms (H, C, N, B) trapped in metal voids.
• Properties: **Non-stoichiometric, hard, high M.P. chemically inert, retain metallic conductivity.**
e.g. TiC, Mn_4N, Fe_3H etc.

2. Standard Electrode Potentials (E^0) *# period m left~right jane m SRP ↑, IE ↑ k wajah se*
• Total $\Delta H = \Delta H_{sub} + IE + \Delta H_{hyd}$. *# ΔH jitna kam uska oxidation utni aasani, utna accha R.A.*
• SRP (Standard Reduction Potential): **Lower SRP = Better Reducing Agent**.
• (Note) **Copper Exception**: Cu is the only 3d metal with positive E^0 (+0.34V). It cannot displace H_2 from dilute acids.
• Exception :-
- Ni Exception: More negative E^0 than expected due to high hydration enthalpy.
- $Mn^{3+} \rightarrow Mn^{2+}$ (E^0 = +1.57V): Mn^{2+} is very stable (d^5).
- $Fe^{3+} \rightarrow Fe^{2+}$ (E^0 = +0.77V): Fe^{3+} is stable (d^5).

Qs: Of the d4 species, Cr^{2+} is strongly reducing while Mn^{3+} is strongly oxidizing, why?
Ans: Cr^{2+} goes to Cr^{3+} after losing an electron and is stable due to filled t2g orbital and Mn^{3+} gains a electron to go to half filled d5 configuration.

3. Catalytic Properties and Color
• Catalysts: Effective due to variable OS, ability to form complexes and surface area (e.g., V_2O_5 in Contact Process, Fe in Haber's).
• Color: **Caused by d-d transitions** (electron jump between split d-orbitals).
• Colorless Ions: d^0 (Sc^{3+}, Ti^{4+}) and d^{10} (Zn^{2+}, Cu^+).
• LMCT (Ligand-to-Metal Charge Transfer): Responsible for color in $KMnO_4$ (Purple) and $K_2Cr_2O_7$ (Orange) despite being d^0.
*# The transition metals form a large number of complex compounds. This is due to the comparatively **smaller sizes** of the metal ions, their **high ionic charges** and the **availability of d orbitals** for bond formation.*

CuF does not exist because of the powerful oxidising nature of F_2.
CuI2 do not exist because of the powerful reducing nature of I_2.

Many copper (I) compounds are unstable in aqueous solution and undergo disproportionation.
$2Cu^+ \rightarrow Cu^{2+} + Cu$
The stability of Cu^{2+} (aq) rather than Cu^+(aq) is due to the much more negative hydration enthalpy of Cu^{2+} (aq) than Cu^+, which more than compensates for the second ionisation enthalpy of Cu.

IMPORTANT COMPOUNDS

1. Oxides of Transition Metals
• Low OS (+1, +2, +3): Basic (e.g., CrO). Exception: Cr_2O_3 is amphoteric.
• Intermediate OS (+4): Amphoteric (e.g., MnO_2).
• High OS (+5, +6, +7): Acidic (e.g., Mn_2O_7, CrO_3).
• (Formula) $Mn_2O_7 + H_2O \rightarrow 2HMnO_4$ (Permanganic Acid)

Zn, Sn, Be, Al, Pb, Ga, Cd	⇒	All oxides are amphoteric in nature.
As^{3+},Fe^{3+},Sb^{3+},Cr^{3+},V^{5+},Te^{4+}	⇒	Oxides are amphoteric only in these oxidation states.
CO, NO, N_2O	⇒	Only these three oxides are neutral in nature.

2. Potassium Permanganate ($KMnO_4$)
• Prepared from Pyrolusite (MnO_2).
• (Formula) Step 1: $MnO_2 + 2KOH + 1/2\ O_2 \rightarrow K_2MnO_4$ (Green) + H_2O
• (Formula) Step 2:
Disproportionation: (In acidic, neutral or weakly basic medium)
$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$ {Can't be used, traces MnO_2}
Commertial: (Electolytic oxidation)
$MnO_4^{2-} \rightarrow MnO_4^-$ (Anode) + H_2 (Cathode)
Another method:
$MnO_4^{2-} + Cl_2/O_3 \rightarrow MnO_4^- + KCl/O_2$
• Structure: MnO_4^- is Purple, Tetrahedral, Diamagnetic.
• Structure: MnO_4^{2-} is Green, Tetrahedral, Paramagnetic.
• **Thermal Decomposition**: $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$ *#same reaction for sunlight decompostion*
• Reaction with H_2SO_4 (Cold): $KMnO_4 + H_2SO_4 \rightarrow Mn_2O_7 + KHSO_4 + H_2O$ (Explosive).

• Oxidizing n-factors:
    • Acidic: n = 5 ($Mn^{7+} \rightarrow Mn^{2+}$)
    • Neutral/Faintly Alkaline: n = 3 ($Mn^{7+} \rightarrow MnO_2$)
    • Strongly Alkaline: n = 1 ($Mn^{7+} \rightarrow MnO_4^{2-}$)
    • (Important) Iodide reaction: Acidic ($I^- \rightarrow I_2$); Neutral/Base ($I^- \rightarrow IO_3^-$).
It forms dark purple needle like crystals with a metallic lustre. It's melting point is 553K. It is moderately soluble in water giving a purple solution

3. Potassium Dichromate ($K_2Cr_2O_7$)
• Prepared from Chromite ore ($FeCr_2O_4$).
    - Step 1: $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4\{Yellow\} + 2Fe_2O_3 + 8CO_2$
    - Step 2: $2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7\{Orange\} + 2Na^+ + H_2O$
    - Step 3: $Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$
• Equilibrium: $2CrO_4^{2-}$ (Yellow) + $2H^+ \rightleftharpoons Cr_2O_7^{2-}$ (Orange) + H_2O.
• Note: $K_2Cr_2O_7$ is a primary standard (non-hygroscopic).
• Oxidizing Action: Only in acidic medium (n = 6).
• **Chromyl Chloride Test**: $K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \rightarrow 2KHSO_4 + 2CrO_2Cl_2$ (Red Vapors) + $3H_2O$.
• **Reaction with H_2O_2**: Forms CrO_5 (Blue Butterfly structure, OS = +6).
• Heating: $4K_2Cr_2O_7 \rightarrow 4K_2CrO_4\{Potassium\ Chromate\} + 2Cr_2O_3\{Chromic\ oxide\} + 3O_2$

F-BLOCK ELEMENTS

1. General Properties
• Lanthanoids (4f): Ce (58) to Lu (71). **Common OS = +3.**
• Actinoids (5f): Th (90) to Lr (103). Radioactive. Variable OS up to +7.
• Configuration: $(n-2)f^{1-14}\ (n-1)d^{0-1}\ ns^2$
• Lanthanoid Contraction: Steady decrease in size due to poor 4f shielding.
• Actinoid Contraction: Greater than Lanthanoid contraction due to even poorer 5f shielding.
Because lanthanum(Ln) closely resembles the lanthanoids and actinium(Ac) actinoids therefore they are discussed with them even though they are d-block elements.

2. Lanthanoids (4f)
• Mnemonic:

Cene	Par	Nadiya	Prem	*ki*	Samai	Eu	Gad-Gad	Tab	Dil	Hua	Aur	Tum	Ab	Lauto
Ce	Pr	Nd	Pm		Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu

• Ce^{4+}: Stable f^0, strong oxidant (wants to reach +3).
• Eu^{2+}: Stable f^7, strong reductant (wants to reach +3).
• Electronic Configuration Exception: Ce, Ge and Lu *# f ka ek electron 5d p chala jata h.*
• Basicity: $La(OH)_3$ (Max) > $Lu(OH)_3$ (Min) due to size decrease.
• Mischmetall: 95% Lanthanoid + 5% Fe.
• Atomic radius decreases with increase in atomic number {left to right}
+2 aur +4 oxidation state bhi dikhata h par jab half-filled ya fully-filled configuration mile sometimes ;-) .

3. Actinoids (5f)
• Small energy gap between 5f, 6d, and 7s leads to many oxidation states.
• Np and Pu show OS from +3 to +7.
• Radius decreases on a period from left to right.

FINAL FORMULA SHEET

GENERAL PROPERTIES

- Configuration: $(n-1)d^{1-10}\ ns^{0-2}$
- Magnetic Moment: $\mu = \sqrt{[n(n+2)]}$ BM
- Radius Shielding: 3d < 4d ≈ 5d (Groups 4-12)

$KMnO_4$ REDOX (n-factors)

- Acidic: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (n=5)
- Neutral/Weak Base: $MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$ (n=3)
- Basic: $MnO_4^- + e^- \rightarrow MnO_4^{2-}$ (n=1)

$K_2Cr_2O_7$ REDOX

- Acidic: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ (n=6)

PREPARATION EQUATIONS

- $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$
- $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$ (Thermal)
- $K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \rightarrow 2KHSO_4 + 2CrO_2Cl_2 + 3H_2O$ (Chromyl Chloride)
- $Cr_2O_7^{2-} + 4H_2O_2 + 2H^+ \rightarrow 2CrO_5 + 5H_2O$

ULTRA-QUICK REVISION

- Pd: Only element with $5s^0$ configuration.
- Zn, Cd, Hg: Not transition elements (full d-orbitals).
- Cu: Only 3d metal with positive SRP (+0.34V).
- Tungsten (W): Highest MP; Mercury (Hg): Lowest MP.
- Lanthanoid Contraction: Causes Zr ≈ Hf size.
- Mn: Maximum oxidation states (+7) in 3d series.
- Ru/Os: Maximum oxidation states (+8) in d-block.
- $KMnO_4$ color: LMCT (not d-d transition).
- CrO_5: Butterfly structure, Cr is in +6 state.
- Mischmetall: Lanthanoid alloy used in magnesium-based flints.
- Actinoid Contraction: Greater than Lanthanoid due to poorer 5f shielding.
- Ce^{4+}: Good oxidizing agent; Eu^{2+}: Good reducing agent.
- $KMnO_4$ in titrations: Use H_2SO_4, not HCl (HCl gets oxidized to Cl_2).
- Interstitial compounds: Non-stoichiometric, hard, retain conductivity.
- $K_2Cr_2O_7$: Primary standard; $Na_2Cr_2O_7$: Hygroscopic (secondary).
- Oxides: High OS = Acidic; Low OS = Basic.
- Mn^{2+} and Fe^{3+}: Extra stable due to d^5 configuration.
- Sc: No variable oxidation states (+3 only).

Write any one consequence of lanthanoid contraction.
- Similar radii of 4d and 5d series elements/ similar properties/ difficulty in separation of lanthanoids

Transition metals exhibit catalytic properties.
- They have the ability to exhibit variable oxidation states/ tendency to form complex compounds/ large surface area.

Transition metals have high enthalpy of atomisation.
- Due to involvement of (n−1) d and ns electrons which results in strong metallic bond and strong interatomic bonding.

Orange colour of Cr2O7^2− ion changes to yellow when treated with an alkali.
- Due to formation of chromate /CrO4^2− ion

E∘ value for Mn3+/Mn2+ couple is highly positive (+1·57 V) as compared to Cr3+/Cr2+.
- Because Mn2+ is more stable due to stable 3 d^5 configuration whereas Cr3+ is more stable due to stable (t_2g)^3 configuration.

Ce (III) is easily oxidised to Ce (IV). Comment.
- Ce(IV) ion has more stable configuration (4f^0) than Ce(III) ion.

Why do E∘ M2+/M values show irregular trend from Vanadium to Zinc?
- Because of irregular values of (ΔiH1+ΔiH2) and sublimation enthalpies.

In aqueous media, which is a stronger reducing agent Cr2+ or Fe2+ and why?
- Cr2+ is stronger reducing agent than Fe2+
Reason: d4→ d3 occurs in case of Cr2+ to Cr3+
But d6→d5 occurs in case of Fe2+ to Fe3+
In a medium (like water) d3 is more stable as compared to d5

Transition metals form large number of interstitial compounds with H,B,C and N .
- H, B, C and N atoms being small in size get trapped inside the crystal lattices of transition metals.

Highest oxidation state of a metal is shown in its oxide or fluoride only. Justify the statement.
- Because fluorine and oxygen are highly electronegative / both are strong oxidising agents.

$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \longrightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$
$2MnO_4^- + 5NO_2^- + 6H^+ \longrightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$
$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$

Write the balanced ionic equations showing the oxidising action of acidified dichromate ($Cr_2O_7^{2-}$) solution with (i) Iron (II) Ion and (ii) tin (II) ion.
- $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$
- $Cr_2O_7^{2-} + 14H^+ + 3Sn^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 3Sn^{4+}$

When $FeCr_2O_4$ is fused with Na_2CO_3 in the presence of air it gives a yellow solution of compound (A). Compound (A) on acidification gives compound (B). Compound (B) on reaction with KCl forms an orange coloured compound (C). An acidified solution of compound (C) oxidises Na_2SO_3 to (D). Identify (A), (B), (C) and (D).
- A=Na_2CrO_4; B=$Na_2Cr_2O_7$; C=$K_2Cr_2O_7$; D=Na_2SO_4

The product of the oxidation of I^- with MnO_4^- in alkaline medium is :
- IO_3^-

Assertion (A) : Cuprous salts are diamagnetic.
Reason (R) : Cuprous ion has completely filled 3d-orbitals.
- Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).